Cover Page

Built Environment Case Competition (BECC) 2023

RECRE-Tech

Team xyz

1. Introduction

Singapore has a steadfast commitment to embracing sustainability across production, consumption, and waste management. Construction and Demolition (C&D) waste constitutes approximately 20% of the total waste stream [1]. Despite an impressive 99% recycling rate, improvements in C&D waste management are crucial to enhance effectiveness and further minimize environmental impacts.

Singapore's heavy reliance on neighboring countries for aggregate supply poses challenges including supply chain vulnerabilities, limited control over imported aggregate quality and negative environmental impact from natural aggregates (NA) use. Interestingly, 70% of demolition waste in Singapore comprises crushed concrete [2], offering a substantial source for producing recycled concrete aggregates (RCA). To prevent aforementioned unfavorable consequences, utilizing RCA is essential to reduce dependence on imported aggregate.

The Samwoh Eco-Green Building demonstrates successful sustainability by using 100% RCA in its structural concrete without compromising integrity or durability [3]. Despite the property differences between RCA and NA, recycled aggregate concrete (RAC) can be formulated with minimal impact on new concrete properties. Continuous monitoring reveals no abnormalities of the building, showcasing the potential of RCA as a significant breakthrough in concrete technology[4].



Fig. 1 Samwoh Eco-Green Building [5]

ML - Machine Learning
RCA - Recycled Concrete Aggregates
RAC - Recycled Aggregate Concrete
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LIBS - Laser-Induced Breakdown Spectroscopy

2. Problem Statement

As viable alternatives to NA, RCA is promising in driving the concrete industry towards sustainability, reducing the need of NA, conserving landfill space, and reducing emissions.

While RCA is commonly used for subbase and non-structural components like eco-concrete, its applications are limited for structures requiring high strength and durability. Contaminants in RCA, such as residual mortar, lead to higher absorption rates, resulting in increased permeability and reduced strength[2].

Furthermore, there are limitations in using a high percentage of RCA in structural applications. In Singapore, approval from Building and Construction Authority (BCA) is necessary for replacing NA with RCA exceeding 10% by mass[2]. Hence, it becomes ineffective as extensive laboratory tests are required in evaluating the suitability of RAC. Moreover, the high variation in properties of various RCA sources complicates concrete upcycling[4].

To address these, the integration of digitalisation is crucial to improve the efficiency in achieving the consistent, predictable and outstanding performance of RCA in structural work.

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3. Solution

It remains a challenge to effectively upcycle end-of-life concrete. To overcome this issue, we propose using laser-induced breakdown spectroscopy (LIBS) analysis and machine learning (ML) for digitizing end-of-life concrete processing. By implementing digital technology - RECRE-Tech, we aim to streamline and optimize the recycling process, enabling more efficient and sustainable utilization of end-of-life concrete.

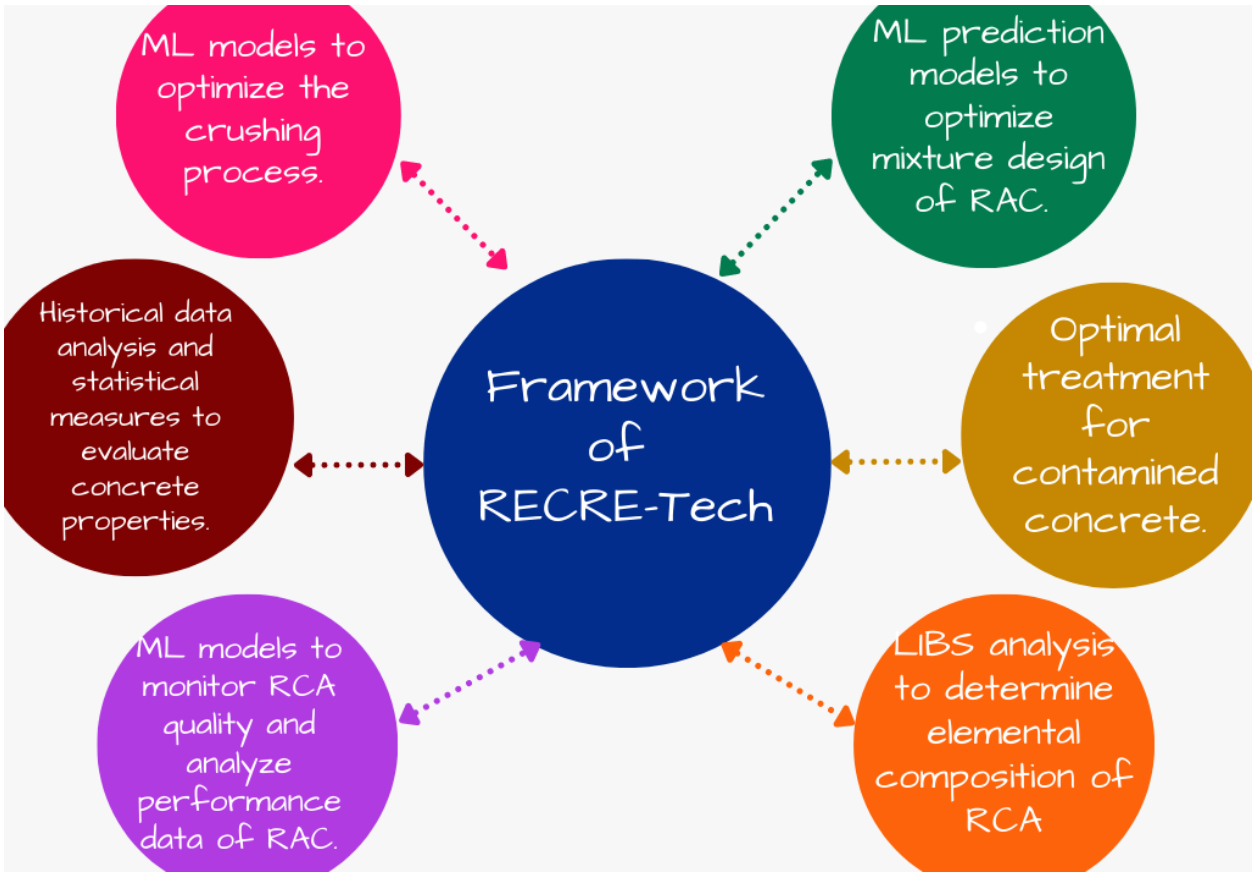


Fig. 2 Framework of RECRE-Tech

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3.1 LIBS analysis for concrete waste processing

Fig. 3 illustrates the typical concrete reclamation process outlined by BCA. With LIBS, a cluster-based identification algorithm to characterize in-line RCA [6], it is prospective that processes ii) and iii) can be made more efficient and streamlined through rapid sorting of recycled concrete particles. LIBS analysis can separate contaminated concrete from those uncontaminated, improving the reliability of RCA in terms of quality.

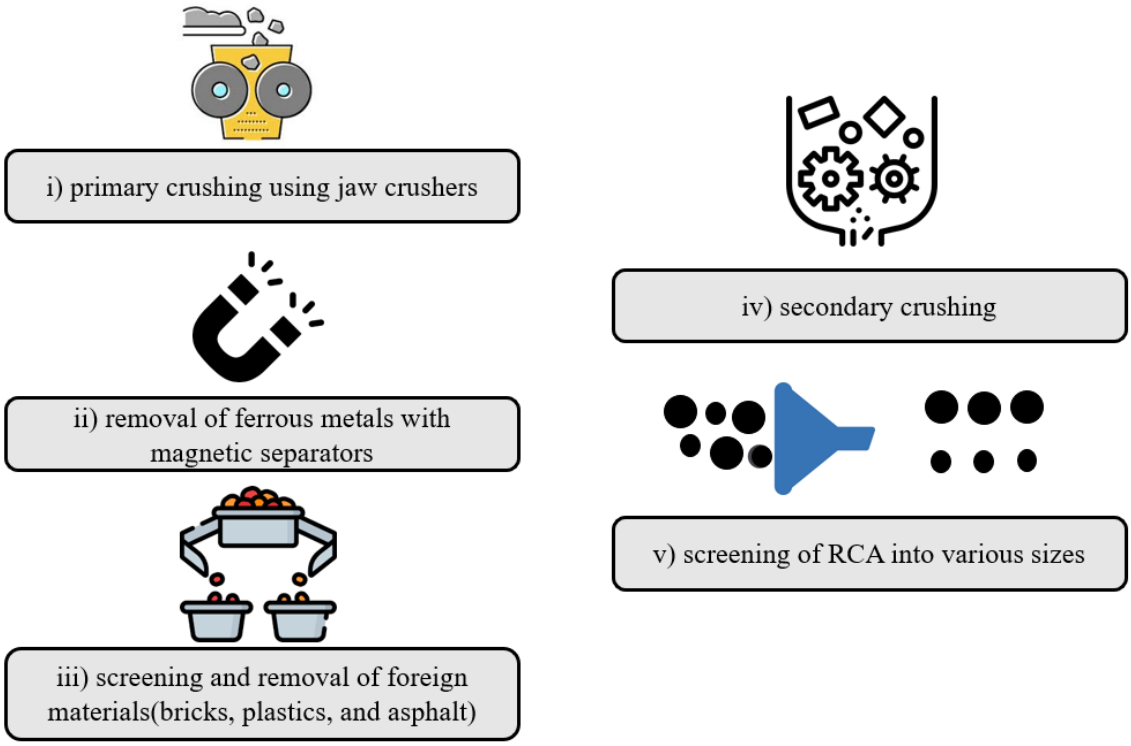


Fig. 3 Typical Concrete Reclamation Process [2]

Our focus lies in digitalizing the concrete waste processing process: integrating sensors into the sorting process to obtain precise data about the amounts of contaminated and uncontaminated RCA, as well as the weight of RCA in different sizes and densities.Throughout the sorting process, sensors identify particles, allowing data to be extracted simultaneously.

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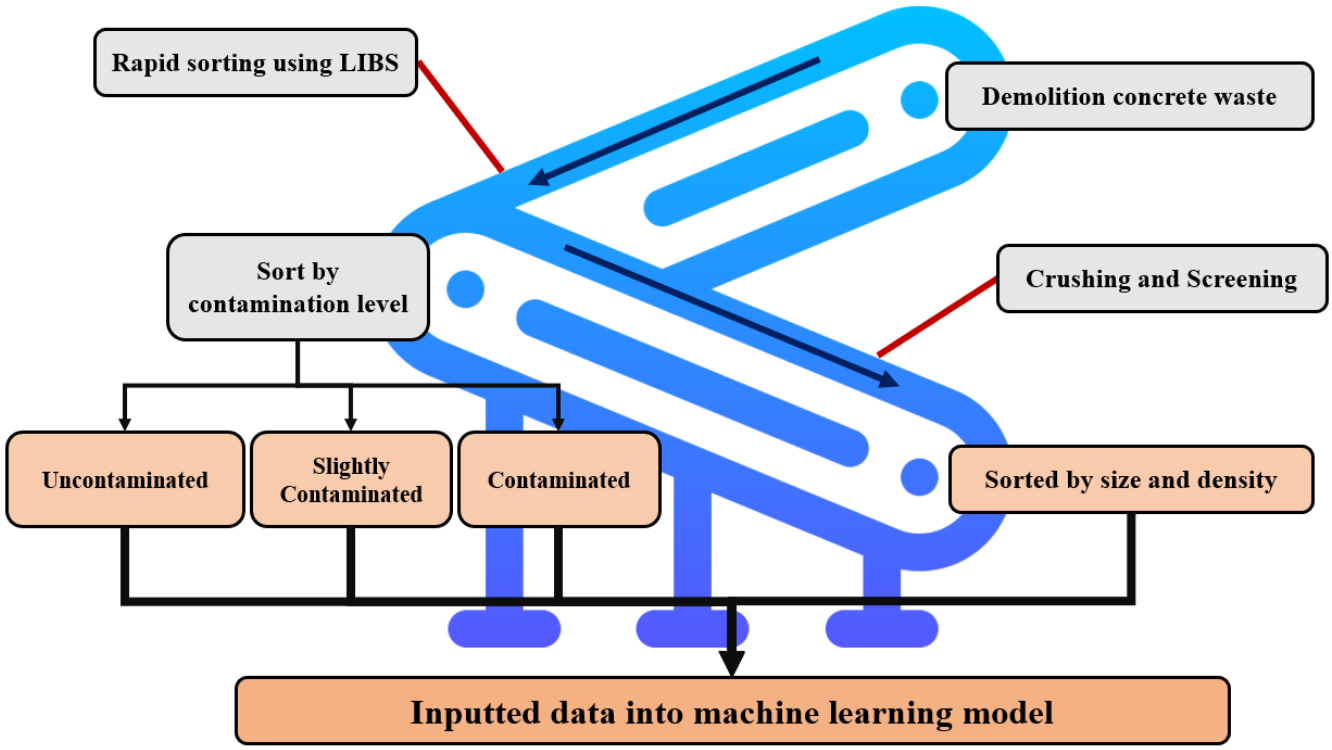


Fig. 4 Processing of demolition concrete waste

3.2 Machine learning (ML)

3.2.1 Optimization of crushing process

As crushing processes contribute to high carbon footprints, its optimisation needs to be achieved using ML. Development of predictive models can estimate system performance based on input factors such as material size, crusher speed and material type [7].

Utilizing historical production data analysis to identify inefficiencies and propose optimization strategies, ML models can help operators in adjusting crusher parameters for optimal performance. By developing real-time control systems for automatic parameter adjustment, consistent output, lower energy consumption, and improved productivity are ensured[8].

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3.2.2 Treatment to improve RCA

To remove the contaminants that adversely affect concrete quality, various treatment methods are available, such as the hybrid chemico-mechanical treatment[9]. Implementing data of contaminated aggregates, ML can help in developing models with parameters as inputs and treatment effectiveness as outputs. As a result, recommendations of the optimal treatment approach for each batch of RCA can be provided.

3.2.3 Production of structural concrete

RECRE-Tech aims to enhance the quality control, standardization, and improvement of RCA for structural concrete. To achieve a mixture optimization, ML prediction models can be developed to predict various RAC properties[10].

As a support to ML models, LIBS analysis determines elemental composition of RCA samples, including contaminants and impurities. Historical data analysis also enables ML to identify patterns and relationships among variables impacting concrete properties. Hence, a large dataset regarding RCA sources and recycling process is required, with validation using a separate dataset to ensure accurate and reliable results[11].

Algorithms like support vector machines, artificial neural networks, decision trees and random forests can be used, while statistical measures evaluate the model performance[12]. A hybrid model combining gradient boosting regression trees model with particle swarm optimization can optimize the mixture design of RAC, meeting compressive strength requirements, reducing cost and mitigating environmental impacts[13].

Additionally, ML models can monitor RCA quality over time and analyze the performance data of RAC structures, hence identifying trends for future design and performance improvement.

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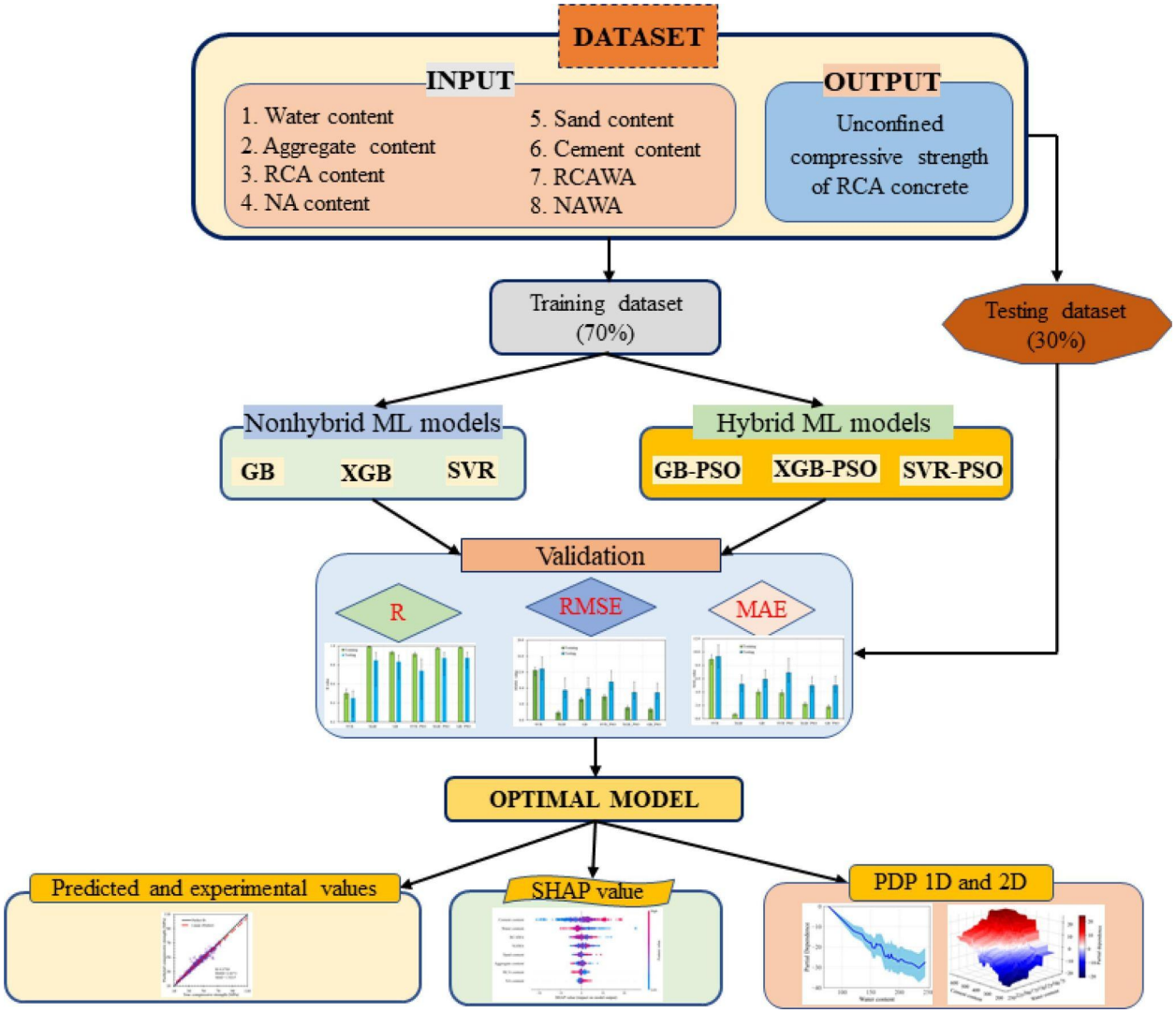


Fig. 5 Flowchart of ML model application for predicting the compressive strength of RCA concrete in this study. [11]

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4. Evaluation - three bottom line analysis

4.1. Profit

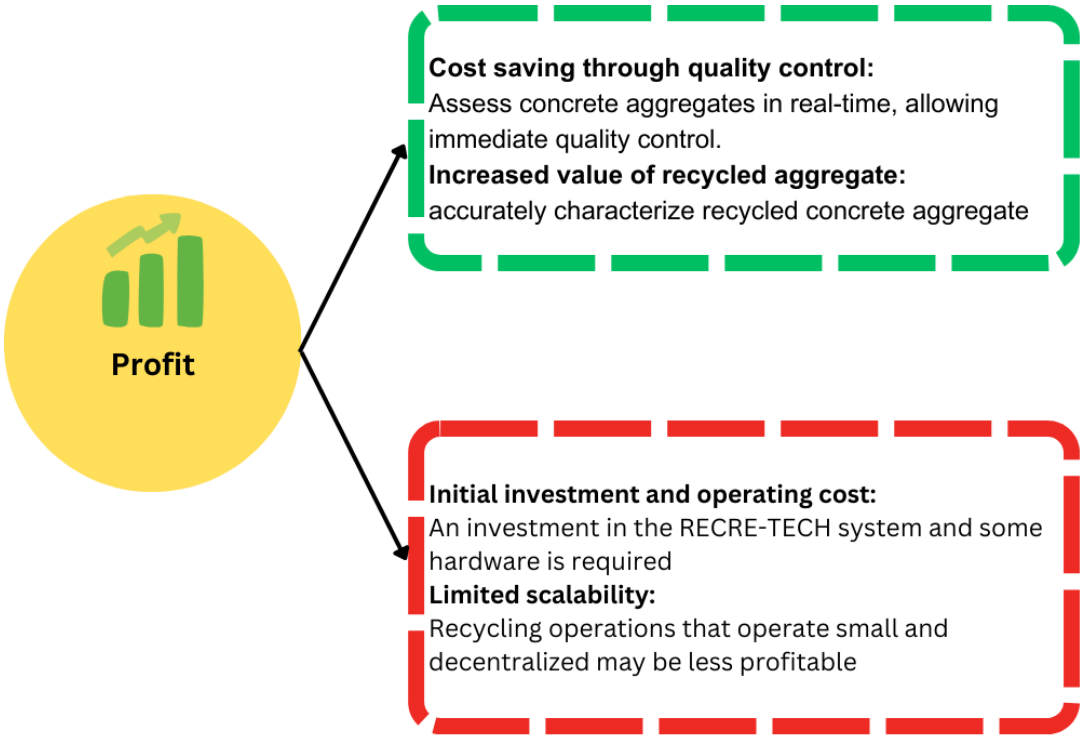


Fig. 6 Strengths and Weaknesses in the Aspect of Profit

RECRE-Tech improves RCA production profitability by identifying aggregate composition issues, reducing rework, and enabling tailored aggregate production for premium marketing. Government and private funding can help overcome high upfront costs, while collaborating with small firms can increase production volumes and cost-effectiveness.

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4.2. Planet

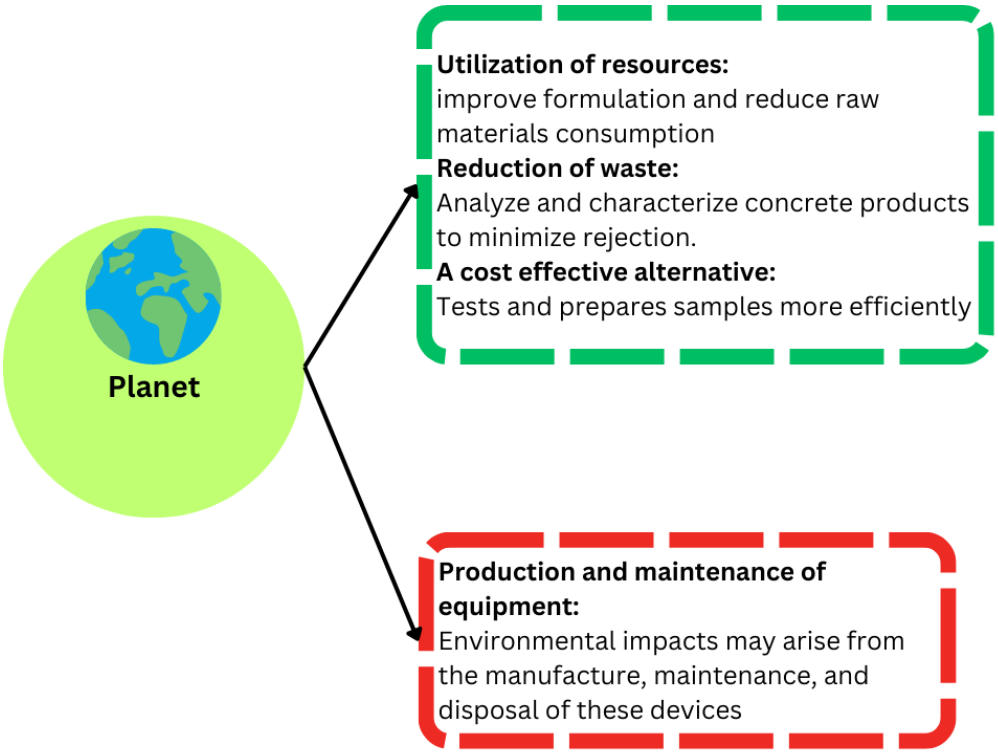


Fig. 7 Strengths and Weaknesses in the Aspect of Planet

RECRE-Tech uses ML to accelerate the analysis of end-of-life concrete, meeting concrete requirements while minimizing NA. Continuous improvement of ML models allows for better concrete design and extended lifespan which promotes environmental sustainability. Manufacturers are encouraged to adopt sustainable practices and improve hardware life-cycle accountability to reduce environmental consequences.

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4.3 People

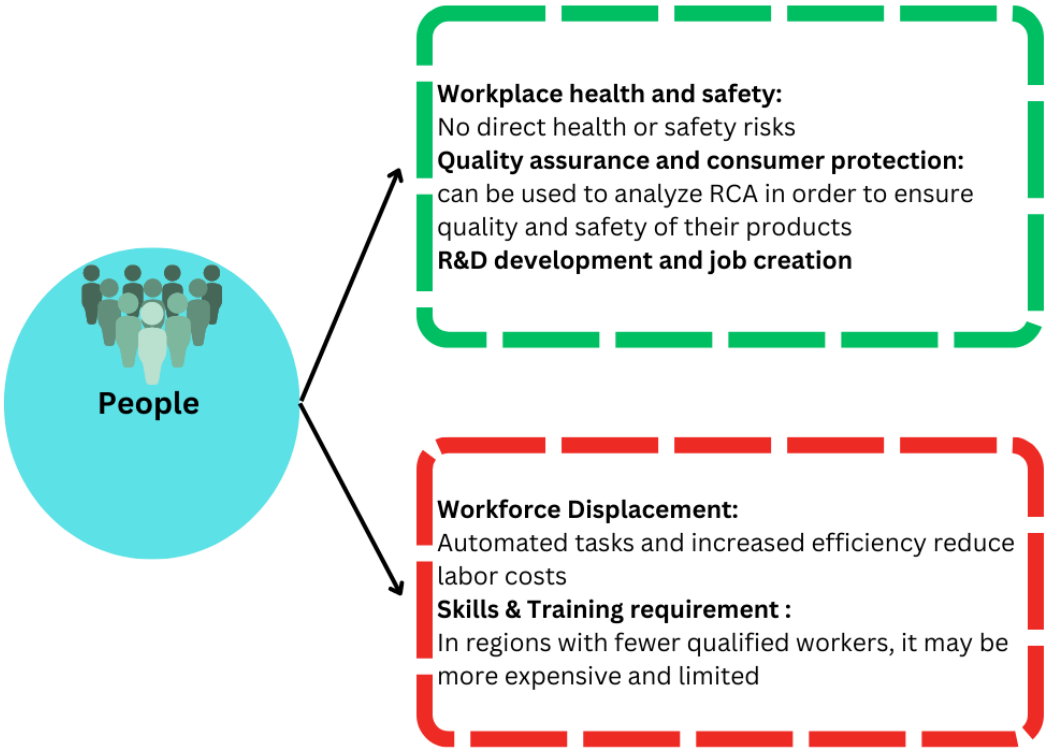


Fig. 8 Strengths and Weaknesses in the Aspect of People

RECRE-Tech enhances workplace safety by minimizing exposure to hazardous materials and creates opportunities for research and development, fostering innovation and expanding career prospects. Training and apprenticeship programs can educate current and potential workers, addressing workplace displacement and the shortage of skilled professionals.

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5. Quantification of Concrete Sustainability

Life Cycle Assessment (LCA) quantifies sustainability of concrete by evaluating its environmental impact throughout its life cycle. To extend beyond environmental factors, sustainability coefficient (k_{TU}) is used to incorporate concrete serviceability and performance.

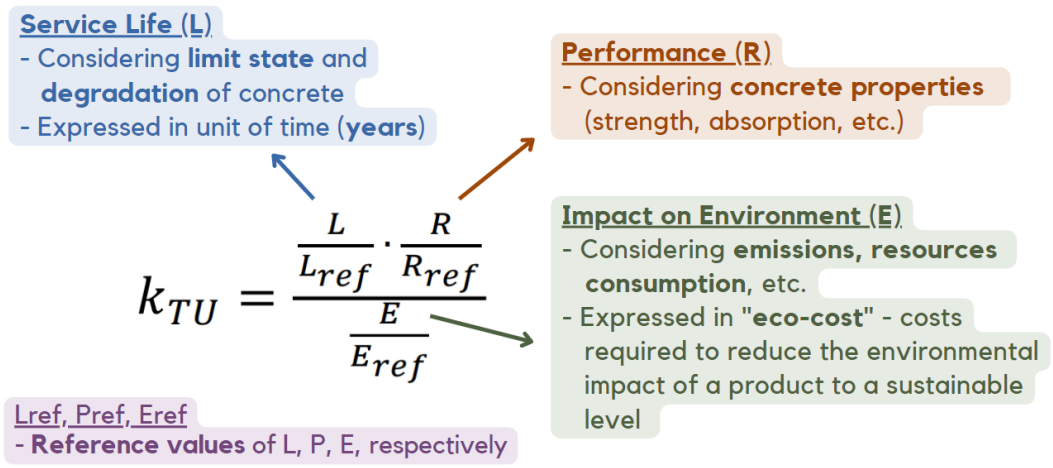


Fig. 9 k_{TU} described by L, R, E and corresponding reference values [14]

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An estimated quantification of concrete sustainability is illustrated as below:

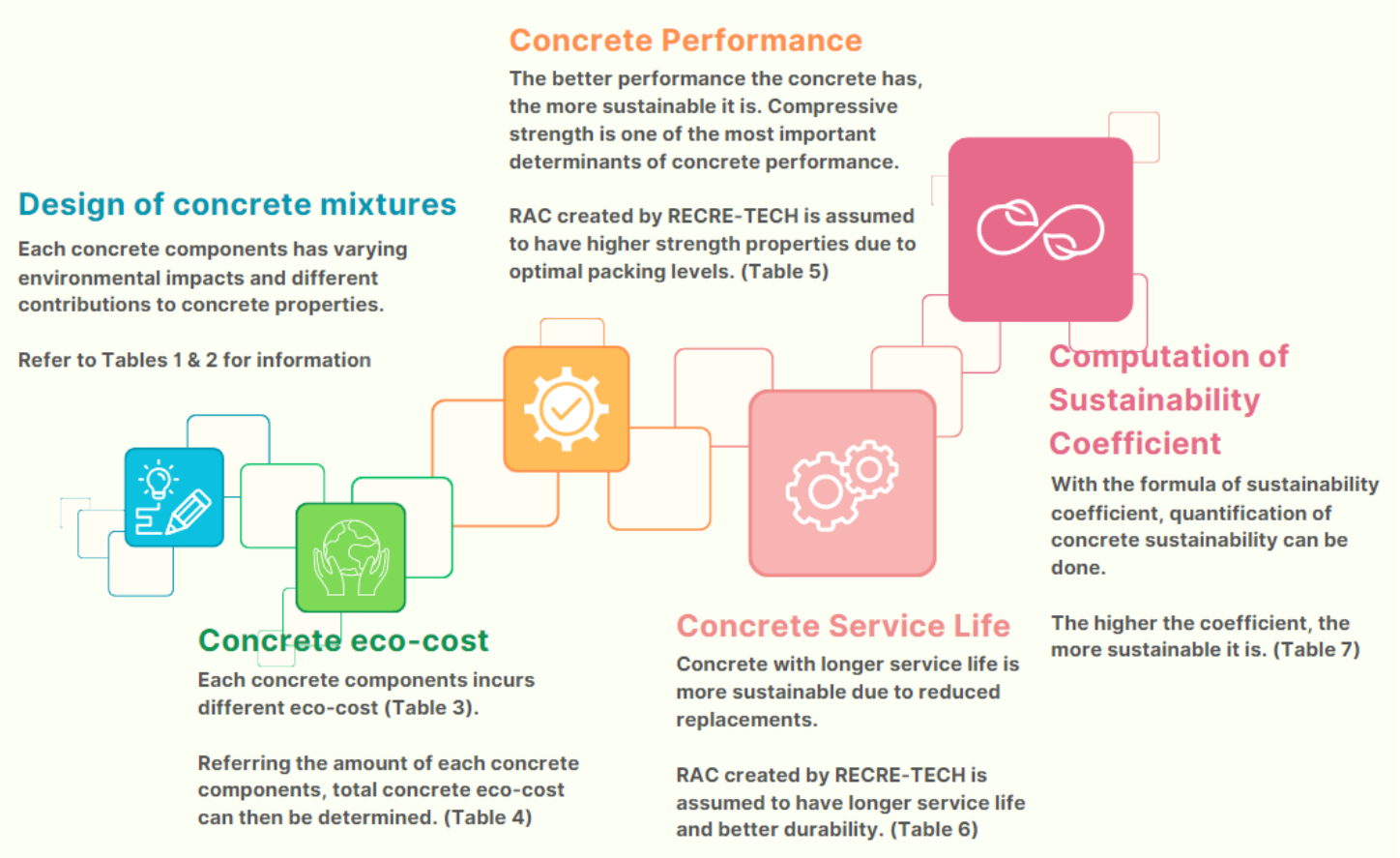


Fig. 10 Steps for Quantification of concrete sustainability

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Table 1 Types of concrete for sustainability quantification

Concrete A	Normal concrete without any RCA replacement, serves as a reference concrete
Concrete B	RAC with 30% of RCA replacement
Concrete C	RAC produced by RECRE-Tech (Improved Concrete B). Assumed to have similar and even better compressive strength and lifespan due to the optimum packing level contributed by improved RCA.

Table 2 Concrete Mixture Proportions (kg/m3)

	Concrete A	Concrete B	Concrete C
NFA	856	856	856
NCA	925	647	647
RCA	0	278	278
Water	169	169	169
Cement	305	305	305

* NFA - Natural Fine Aggregate, NCA - Natural Coarse Aggregate,

* Adapted from “Effect of replacement of 5~13mm recycled coarse aggregates on field” [15]

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Table 3 Environment Cost Per kg of Materials (USD/kg)

	NFA	NCA	RCA	Water	Cement
ADP	8.86E-08	1.85E-07	1.78E-06	3.13E-08	6.30E-05
GWP	1.14E-05	3.33E-05	7.69E-05	5.26E-07	4.87E-03
ODP	5.58E-09	7.75E-10	1.22E-07	1.23E-13	2.32E-07
AP	6.86E-05	1.23E-04	9.31E-04	2.13E-06	1.09E-02
POCP	2.95E-06	5.67E-07	4.76E-05	4.13E-09	1.67E-03
EP	2.12E-06	4.05E-06	2.95E-05	6.45E-08	2.01E-04
Eco-cost of Materials (USD/kg)	**8.51E-05**	**1.61E-04**	**1.09E-03**	**2.75E-06**	**1.77E-02**

* Environmental Impacts include abiotic depletion potential (ADP), global warming potential (GWP), ozone depletion potential (ODP), acidification potential(AP), photochemical ozone creation potential (POCP), eutrophication potential (EP).

* Adapted from "*Environmental evaluation of concrete containing recycled and by-product aggregates based on life cycle assessment* " Table 4,5,6 [16]

Table 4 Environmental Cost per m3 of Each Concrete (USD/m3)

	Concrete A	Concrete B	Concrete C
Total Eco-Cost	**5.62**	**5.88**	**5.88**

* By summing the production of eco-costs and amounts of components, using information in Tables 2&3

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The comparison of eco-cost between RAC and NAC in Table 3 may not accurately reflect the Singapore context. RAC generally has higher eco-costs due to increased energy consumption during recycling. However, Singapore's reliance on NA importation must be considered: higher emissions due to NA transportation from distant locations. Unfortunately, this factor is not included in the available statistics. Hence, actual eco-costs of NAC are expected to be higher than RAC in Singapore.

Table 5 56-day Compressive Strength of Concretes under Standard Curing (MPa)

	Concrete A	Concrete B	Concrete C
56-day Compressive Strength (MPa)	52	51	53

* Adapted from "Effect of replacement of 5~13mm recycled coarse aggregates on field" [15]

Table 6 Service Life of Concretes (years)

	Concrete A	Concrete B	Concrete C
Service Life (L)	100	93	97

* Estimated from "*Estimated service life of ordinary and high-performance reinforced recycled aggregate concrete*" [17]

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Table 7 Sustainability Coefficient

	Concrete B	Concrete C
Environment (E/E_{ref})	5.88/5.62	5.88/5.62
Performancet (R/R_{ref})	51/52	53/52
Service Life (L/L_{ref}))	93/100	97/100
Sustainability Coefficient	0.87	0.94

* Calculated using formula in Fig. 9

* Values of Concrete A served as reference values

Table 7 indicates that RECRE-Tech will successfully increase sustainability by reducing environmental impact, improving performance and service life. Nevertheless, k_{TU} of both concretes are expected to be even higher due to the reduced E/E_{ref} in Singapore's case.

6. Conclusion

RECRE-Tech leverages ML and LIBS analysis to streamline RCA production and improve RAC quality. This sustainable solution optimizes end-of-life concrete management by reducing the environmental impact while promoting economic growth and social well-being.

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Team xyz

RECRE-Tech

- Recycled Concrete Technology -
A transition to Recycled Concrete Aggregate (RCA)

THE PROBLEMS

- **Undesirable properties** of RCA (low strength, durability)
- **High variation** of RCA properties from different sources
- **Limited use** of RCA in structural work
- **Complicated recycling process**
- **Extensive laboratory test** to access suitability of concrete

WHAT IS RECRE-Tech?

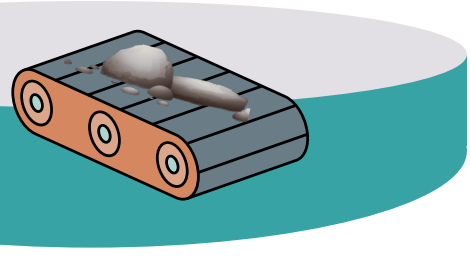
A model that aims to

- reduce the environmental impact of concrete production
- improve the efficiency in RCA handling process
- improve the quality of RCA and widen its use in structural work
- predict concrete properties

DEVELOPMENT OF A MACHINE LEARNING MODEL TO PREDICT RCA PERFORMACE

To achieve **quality control, standardization, and improvement** of RCA for use as structural concrete.

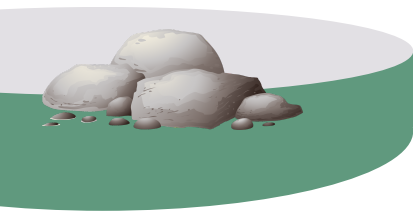
HOW DOES IT WORK?



01

Sorting

Efficient sorting with LIBS technology to separate contaminated aggregates



02

Crushing and Screening

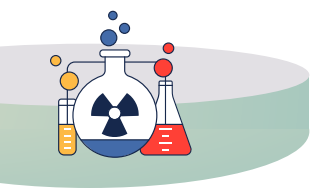
Optimized crushing and screening process for different batches of RCA



03

Data Processing

Provide data (contamination/size/ density) for Machine Learning model



04

RCA Treatment

Treatment of recycled concrete aggregate based on their contamination level

OPTIMIZATION OF CRUSHING PROCESS

Using machine learning to achieve real-time control system for lower energy consumption and better productivity

TREATMENT TO IMPROVE RCA

Implementing data of contaminated aggregates to recommend optimal treatment approaches

CONCRETE MIX DESIGN & PROPERTIES PREDICTION

Using historical data analysis and algorithms to optimize the mixture design of RAC for specific strength requirements

